

Antique Street Lights (streetlights)

George is very passionate about the *good old times*, so much that he switched the illumination in his street to old-fashioned oil lanterns.

The street is now nicely illuminated by a sequence of N equally spaced lanterns, providing joy and delight to the whole neighbourhood.

However, he got the permission for doing so only by accepting to personally maintain the lanterns lit up: in fact, he has to ensure that given any subsequence of consecutive M lanterns, *at least* K of them are lit up at any given time.

Tonight an heavy storm is coming down on George's neighbourhood, and some particularly strong wind gusts just extinguished part of the lanterns. George can see for each lantern i whether it is lit ($L_i = 1$) or not ($L_i = 0$).

Since he doesn't want to risk his concession to be revoked, it is time to get out in the storm and turn on some lights!



Figure 1: An artistic view of George's street.

Help George not to freeze in the storm, and find the minimum amount of lanterns which need to be lit up in order to comply with the city regulations!

 Among the attachments of this task you may find a template file `streetlights.*` with a sample incomplete implementation.

Input

The first line contains three integers N , M , K .

The second line contains N integers L_i , such that $L_i = 0$ if the corresponding lantern is off and $L_i = 1$ if it is on.

Output

You need to write a single line with an integer: the minimum number of lanterns which need to be lit up.

Constraints

- $1 \leq K \leq M \leq N \leq 100\,000$.
- $0 \leq L_i \leq 1$ for each $i = 0 \dots N - 1$.



Scoring

Your program will be tested against several test cases grouped in subtasks. In order to obtain the score of a subtask, your program needs to correctly solve all of its test cases.

- **Subtask 1** [5 points]: Examples.
- **Subtask 2** [10 points]: $K = M$.
- **Subtask 3** [10 points]: $M = N$.
- **Subtask 4** [20 points]: $N \leq 20$.
- **Subtask 5** [25 points]: $N \leq 1000$.
- **Subtask 6** [30 points]: No additional limitations.

Examples

input.txt	output.txt
5 2 1 0 1 0 0 1	1
12 4 2 0 0 0 0 1 0 1 0 0 0 1 0	3

Explanation

In the **first sample case**, it is sufficient to lit up the lantern with $i = 2$ obtaining the following configuration:

0 1 1 0 1

In the **second sample case**, it is sufficient to lit up the lanterns with $i = 2, 3, 8$ obtaining the following configuration:

0 0 1 1 1 0 1 0 1 0 1 0